

ANDREW ALAGNA

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SOFTWARE ENGINEER | FRONTEND & FULL-STACK DEVELOPMENT | ACCESSIBILITY & PERFORMANCE

Software Engineer with 3+ years building high-performance, WCAG AAA-compliant web applications at American Express. Specialized in React.js and Node.js with proven expertise in accessibility, performance optimization, and A/B testing that delivered measurable business impact for 50M+ users across 10+ international markets.

PROFESSIONAL EXPERIENCE

American Express

Software Engineer | 08/2022 - Present

- Spearheaded Web Content Accessibility Guidelines (WCAG) AAA compliance implementation across 10+ international markets using React.js, HTML5, CSS3, and JavaScript, improving accessibility audit scores from 72% to 99%, mitigating legal risk and expanding market reach for 30M+ global users
- Engineered secure user account management system using React.js, Node.js, and Kotlin, enabling 5M+ users to update sensitive information (name, address, contact details, income) while maintaining data integrity and regulatory compliance
- Optimized React component rendering and Backend for Frontend (BFF) API architecture, reducing average page load time by 10% (from 3.2s to 2.9s) across consumer, small business, and corporate products, improving experience for 50M+ active users
- Designed and executed 15+ A/B experiments using JavaScript and analytics APIs, increasing form completion rates by 12% and customer satisfaction scores by 15% based on 50K+ user feedback responses
- Built automated Continuous Integration/Continuous Deployment (CI/CD) pipelines using GitHub Actions, reducing deployment time by 15% and implementing feature flagging system with real-time toggles for safer releases
- Collaborated in Agile environment to deliver enterprise-scale applications, implementing comprehensive Test-Driven Development (TDD) strategy with Jest/React Testing Library, achieving 100% code coverage and reducing production bugs by 40%

CodePath

Intro to Software Engineering - Teaching Assistant | 05/2022 - 08/2022

- Led weekly instruction sessions for 180+ students in Python-based Data Structures & Algorithms, achieving 95% student satisfaction rating through clear explanations and engaging problem-solving exercises
- Designed 20+ coding exercises and debugging workshops that improved student technical interview performance, demonstrating strong communication and curriculum development skills

- Mentored students on technical concepts and career readiness, contributing to measurable improvements in their confidence and job search outcomes

New York City Department of Transportation

Geographic Information Systems Intern | 04/2022 - 08/2022

- Engineered Python application to optimize processing and visualization of NYC's census data, improving data accuracy and accessibility for city-wide analysis and planning initiatives
 - Streamlined data organization and labeling processes using Python automation, reducing processing time from weeks to seconds for enhanced geospatial mapping workflows
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VOLUNTEER LEADERSHIP

CodePath | 06/2021 - Present

Teaching & Mentoring

- Mentored and instructed 200+ students across multiple cohorts in Data Structures & Algorithms, cybersecurity fundamentals, and career readiness, fostering an inclusive learning environment
 - Developed and presented engaging technical curriculum that contributed to 90%+ student satisfaction in end-of-course surveys, demonstrating strong communication and teaching skills
 - Guided students weekly through hands-on debugging, problem-solving, and technical interview preparation, securing free professional certifications for 100+ students
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EDUCATION

Bachelor of Arts in Computer Science

Hunter College | New York, NY | Graduated: May 2022 | GPA: 3.62, Cum Laude

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, PostgreSQL, SQLite

Frontend: React.js, Redux, HTML5, CSS3, Tailwind CSS, Component Libraries, Responsive Design, Single Page Applications

Backend & APIs: Node.js, Kotlin, GraphQL, REST APIs, BFF Architecture, Microservices

DevOps & Tools: AWS, Docker, Git, GitHub Actions (CI/CD), Jenkins

Testing & Quality: Jest, React Testing Library, Test-Driven Development (TDD)

Core Expertise: WCAG AAA Accessibility, Agile/Scrum Methodologies, Performance Optimization, System Design, A/B Testing, State Management